Chapter 4 solutions

CH04-E1

1. Solve

$$x_1 + 2x_2 = 0, \qquad x_2 + x_3 = 0.$$

Set $x_2 = t$, giving $x_1 = -2t, x_3 = -t$. Hence

$$N(A) = \operatorname{span}\{(-2, 1, -1)^T\}.$$

2. The independent rows $r_1 = (1, 2, 0)^T, r_2 = (0, 1, 1)^T$ form a row-space basis. For $n = (-2, 1, -1)^T$,

$$r_1^T n = -2 + 2 = 0, \qquad r_2^T n = 1 - 1 = 0.$$

Every row-space combination is therefore perpendicular to n. The orthogonal spaces have dimensions 2 and 1, summing to 3, so they are orthogonal complements.

3. The 2×3 matrix has two independent rows and rank 2. Its column space is a two-dimensional subspace of $\mathbb{R}^2$, hence all of $\mathbb{R}^2$. The left nullity is

$$\dim N(A^T) = m - r = 2 - 2 = 0,$$

Thus $N(A^T) = \{0\}$.

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CH04-E2

1. Since $a^T a = 3$ and $a^T b = 7$,

$$p = \frac{7}{3} a = \frac{7}{3}(1, 1, 1)^T.$$

2.

$$e = b - p = \left(2 - \frac{7}{3}, 1 - \frac{7}{3}, 4 - \frac{7}{3}\right)^T = \frac{1}{3}(-1, -4, 5)^T.$$

3. $a^T e = (-1 - 4 + 5)/3 = 0$, proving orthogonality to the line.

4. The line projector is

$$P = \frac{a a^T}{a^T a} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

It is symmetric. The product $J_3 J_3$ has every entry equal to 3, so

$$P^2 = \frac{1}{3^2} \cdot 3\, J_3 = \frac{1}{3} J_3 = P,$$

where J_3 is the all-ones matrix.

Chapter 4 solutions (continued)

CH04-E3

1.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} c \\ m \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

2.

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

The equations are

$$3c + 3m = 5, \qquad 3c + 5m = 6.$$

Subtraction gives $2m = 1$, hence $m = 1/2$, and substitution gives $c = 7/6$.

3.

$$p = Ax = \left(\frac{7}{6}, \frac{7}{6} + \frac{3}{6}, \frac{7}{6} + \frac{6}{6}\right)^T = \left(\frac{7}{6}, \frac{5}{3}, \frac{13}{6}\right)^T.$$

$$e = b - p = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right)^T.$$

The residual sum is zero, establishing orthogonality to the constant column. Also

$$t \cdot e = 0 + \frac{1}{3} + \left(-\frac{1}{3}\right) = 0.$$

Hence $A^T e = 0$.

4. $\|e\|^2 = 2/36 + 4/36 = 1/6$, the minimum sum of squared residuals over all c, m.

CH04-E4

1. Since $A^T A$ is symmetric, its inverse is symmetric.

$$P^T = A((A^T A)^{-1})^T A^T = A(A^T A)^{-1} A^T = P.$$

Also

$$P^2 = A(A^T A)^{-1} A^T A(A^T A)^{-1} A^T = P.$$

2. The complement is symmetric and

$$(I - P)^2 = I - 2P + P^2 = I - P.$$

Thus it is an orthogonal projector. The cross product is

$$P(I - P) = P - P^2 = 0.$$

Its range is perpendicular to the range of P because $P^T(I - P) = 0$.

Chapter 4 solutions (continued)

3. If $A^TAz = 0$, multiply by z^T to get

$$0 = z^TA^TAz = \|Az\|^2.$$

Therefore $Az = 0$. Column independence gives $z = 0$, proving a trivial nullspace and invertibility of A^TA.

4. Dependent columns give nonzero z with $Az = 0$, hence

$$A^TAz = 0,$$

This nonzero nullspace vector proves singularity of A^TA.

CH04-E5

1. The norm of a_1 is $\sqrt{2}$, so

$$q_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)^T.$$

Since $q_1^Ta_2 = 1/\sqrt{2}$,

$$z_2 = a_2 - \frac{1}{2}(1,1,0)^T = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)^T.$$

$$\|z_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}},$$

$$q_2 = \frac{1}{\sqrt{6}}(1,-1,2)^T.$$

2. The Gram-Schmidt representations are

$$a_1 = \|a_1\|q_1 = \sqrt{2}\,q_1,$$

$$a_2 = \frac{1}{\sqrt{2}}q_1 + \sqrt{\frac{3}{2}}q_2.$$

Therefore

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

3. Both columns have unit norm, and

$$q_1^Tq_2 = \frac{1 + (-1) + 0}{\sqrt{12}} = 0,$$

Thus $Q^TQ = I_2$. Expanding by columns gives

$$\sqrt{2}q_1 = a_1,$$

$$\frac{1}{\sqrt{2}}q_1 + \sqrt{\frac{3}{2}}q_2 = \left(\frac{1}{2}, \frac{1}{2}, 0\right)^T + \left(\frac{1}{2}, -\frac{1}{2}, 1\right)^T = a_2,$$

Chapter 4 solutions (continued)

Hence $QR = A$.

CH04-E6

1.

$$Q^T b = \left(\frac{1+2}{\sqrt{2}}, \frac{1-2}{\sqrt{6}}\right)^T = \left(\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{6}}\right)^T.$$

Solve

$$\sqrt{2}\,x_1 + \frac{1}{\sqrt{2}}x_2 = \frac{3}{\sqrt{2}}, \qquad \sqrt{\frac{3}{2}}x_2 = -\frac{1}{\sqrt{6}}.$$

The second equation gives $x_2 = -1/3$; the first then gives $x_1 = 5/3$. Thus

$$x = \begin{bmatrix} 5/3 \\ -1/3 \end{bmatrix}.$$

2.

$$p = Ax = \left(\frac{4}{3}, \frac{5}{3}, -\frac{1}{3}\right)^T,$$

$$e = b - p = \frac{1}{3}(-1, 1, 1)^T, \qquad \|e\|^2 = \frac{1}{3}.$$

3.

$$a_1^T e = -\frac{1}{3} + \frac{1}{3} = 0,$$

$$a_2^T e = -\frac{1}{3} + \frac{1}{3} = 0.$$

This nonzero residual distinguishes a least-squares solution from an exact solution. Columnwise orthogonality is precisely $A^T e = 0$, and full column rank guarantees a unique parameter vector.

CH04-E7

1. The columns are identical, so they are dependent and rank is 1.
2. Set $a = (1, 2)^T$. Then

$$a^T a = 5, \qquad a^T b = 1 + 6 = 7,$$

$$p = \frac{7}{5}a = \left(\frac{7}{5}, \frac{14}{5}\right)^T.$$

3.

$$A^T A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 7 \\ 7 \end{bmatrix}.$$

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Chapter 4 solutions (continued)

Every minimizer has the form

$$x^\star = \begin{bmatrix} 7/5 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

4. Differences of parameters producing p lie in $N(A) = \operatorname{span}\{(-1,1)^T\}$. This nonzero direction gives nonunique parameters, while all map to the same nearest column-space point p.

CH04-E8

1. Direct multiplication gives

$$P^2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = P.$$

2. However,

$$P^T = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \neq P.$$

3. Since $P(x,y)^T = (x+y,0)^T$,

$$R(P) = \operatorname{span}\{(1,0)^T\}, \qquad N(P) = \operatorname{span}\{(-1,1)^T\}.$$

4. $R(P)$ and $N(P)$ are not orthogonal, since $(1,0)^T \cdot (-1,1)^T = -1 \neq 0$. Thus $P^2 = P$ describes an oblique projector; orthogonal projection also requires $P^T = P$.

CH04-E9

The row space has basis $(1,1,0)^T$, the column space $(1,2)^T$, both of dimension 1. Solving Ax=0 gives basis $(1,-1,0)^T, (0,0,1)^T$, dimension 2. The left nullspace has basis $(2,-1)^T$, dimension 1. Their dot products verify the two orthogonal pairs, and dimensions sum to 3 on the input side and 2 on the output side.

Projection onto c=(1,2) gives $p = c(c^T b)/(c^T c) = (1,2)^T$, $e = (2,-1)^T$, with cᵀe=0. For any x, e is orthogonal to p−Ax, so

$$\|b - Ax\|^2 = 5 + \|p - Ax\|^2 \geq 5.$$

Equality holds exactly when Ax=p, or $x_1+x_2=1$. Hence all minimizers are $x = (1/2 + t, 1/2 - t, s)^T$. Split this into $\hat{x} = (1/2, 1/2, 0)^T$ and $z = (t, -t, s)^T$, in the orthogonal row and null spaces. Thus $\|x\|^2 = 1/2 + 2t^2 + s^2$, uniquely minimized at t=s=0, with norm $1/\sqrt{2}$.

Chapter 4 solutions (continued)

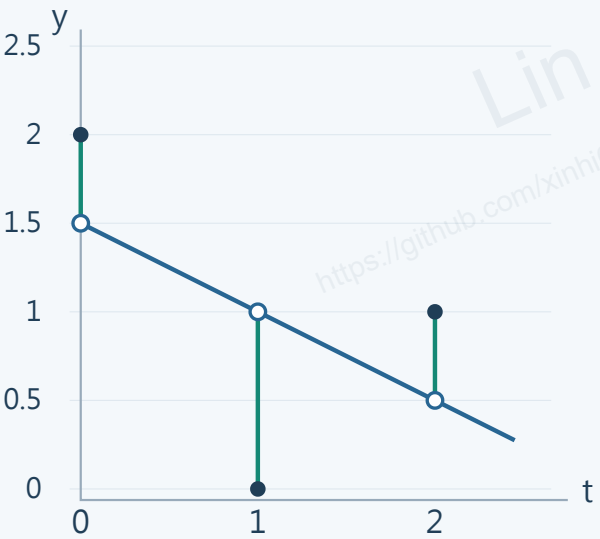
For the particular solution, $(2,-1,2)^T = \hat{x} + (3/2,-3/2,2)^T$. Its nullspace component maps to zero and has squared norm 17/2. Its total squared norm is 1/2+17/2=9, so its norm is 3. It has the same projected output p and minimum residual norm $\sqrt{5}$, but a larger parameter norm. Residual norm and parameter norm are distinct objectives.

CH04-E10

The design matrix remains $A = [(1,1,1)^T\ (0,1,2)^T]$, but $b = (2,0,1)^T$. The normal equations are $3c + 3m = 3$, $3c + 5m = 2$. Subtraction gives $2m = -1$, hence $m = -1/2, c = 3/2$.

Time t	Observed	Predicted	Residual	Squared residual
0	2	$3/2$	$1/2$	$1/4$
1	0	1	-1	1
2	1	$1/2$	$1/2$	$1/4$

Hold t fixed: measure three vertical residuals



Filled dots: observations b_i

Open dots: predictions p_i

Blue line: y = 3/2 − t/2

Green: $e_i = b_i - p_i$

$e = (1/2, -1, 1/2)^T$

$\Sigma\, e_i^2 = 3/2$

1/2 − 1 + 1/2 = 0

0(1/2) + 1(−1) + 2(1/2) = 0

Equal units on t and y; every green segment holds t fixed.

The residual is $e = (1/2,-1,1/2)^T$, with $\|e\|^2 = 3/2$. The checks $1/2 - 1 + 1/2 = 0$ and $-1 + 2(1/2) = 0$ establish $A^T e = 0$.

Since $A^T e = 0$, every $h = (\Delta c, \Delta m)^T$ satisfies

$$\|b - A(x^\star + h)\|^2 = \frac{3}{2} + \|Ah\|^2 = \frac{3}{2} + 3(\Delta c + \Delta m)^2 + 2(\Delta m)^2.$$

Chapter 4 solutions (continued)

The increment vanishes only at $h = 0$, proving uniqueness. A zero sum checks only the first column: $d = (1, -1, 0)^T$ sums to zero but has inner product -1 with the time column.

After centering, the orthogonal columns have right-hand-side inner products 3 and −1. Hence $a = 3/3 = 1, m = -1/2, c = a - m = 3/2$. Thin QR uses the same Q, R as the lesson, but now

$$Q^T b = \begin{bmatrix} \sqrt{3} \\ -1/\sqrt{2} \end{bmatrix}, \qquad q_3^T b = \frac{3}{\sqrt{6}}, \qquad (q_3^T b)^2 = \frac{3}{2}.$$

The triangular equations give $\sqrt{2}m = -1/\sqrt{2}$ and $\sqrt{3}(c + m) = \sqrt{3}$, confirming the normal-equation result. Plot segments hold time fixed and compare heights. After combining observations into a three-dimensional vector, e is perpendicular to the entire reachable prediction space $C(A)$. The two drawings have different coordinate spaces and different objects to which perpendicularity refers.

Verification code appendix: projection_and_qr.py

```
"""Check the line projection and Gram-Schmidt example in chapter four."""

from math import sqrt


def dot(left, right):
    return sum(x * y for x, y in zip(left, right))


def scale(value, vector):
    return [value * x for x in vector]


def add(left, right):
    return [x + y for x, y in zip(left, right)]


def norm(vector):
    return sqrt(dot(vector, vector))


a = [1.0, 1.0, 1.0]
b = [2.0, 1.0, 4.0]
p = scale(dot(a, b) / dot(a, a), a)
e = add(b, scale(-1.0, p))
assert abs(dot(a, e)) < 1e-12
assert abs(norm(e) - sqrt(42) / 3) < 1e-12

a1 = [1.0, 1.0, 0.0]
```

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Chapter 4 solutions (continued)

```
a2 = [1.0, 0.0, 1.0]
q1 = scale(1.0 / norm(a1), a1)
z2 = add(a2, scale(-dot(q1, a2), q1))
q2 = scale(1.0 / norm(z2), z2)
assert abs(dot(q1, q1) - 1.0) < 1e-12
assert abs(dot(q2, q2) - 1.0) < 1e-12
assert abs(dot(q1, q2)) < 1e-12

r11 = sqrt(2.0)
r12 = 1.0 / sqrt(2.0)
r22 = sqrt(1.5)
reconstructed_a1 = scale(r11, q1)
reconstructed_a2 = add(scale(r12, q1), scale(r22, q2))
assert all(abs(x - y) < 1e-12 for x, y in zip(reconstructed_a1, a1))
assert all(abs(x - y) < 1e-12 for x, y in zip(reconstructed_a2, a2))

target = a2
qtb = [dot(q1, target), dot(q2, target)]
x2 = qtb[1] / r22
x1 = (qtb[0] - r12 * x2) / r11
fit = add(scale(x1, a1), scale(x2, a2))
assert abs(x1) < 1e-12 and abs(x2 - 1.0) < 1e-12
assert norm(add(target, scale(-1.0, fit))) < 1e-12

print("line projection residual norm =", norm(e))
print("q1 =", q1)
print("q2 =", q2)
```

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References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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